



Fig. 1 | Correlative 3D microscopy, in situ mechanics and experimentally informed simulations to reveal the mechanical deformation behavior of complex 3D nanosponges on two different length scales. The 3D network structure of FIB prepared npg pillars is obtained by (1) non-destructive tomography and used as input for (2) correlative mechanical simulations. After (3) experimental compression the 3D network is (4) visualized again by means of the tomographic techniques. The resulting experimental and simulated stress-strain response, strain fields and crystal defects are compared and evaluated. For small ligament length ($L \sim 30$ nm) 360° electron tomography (ET) is combined with in situ TEM compression testing and molecular dynamics (MD) simulations, while on larger length scale ($L \sim 300$ nm) nano X-ray computed tomography (nano-CT) is combined with finite element (FE) crystal plasticity simulations and in situ scanning electron microscopy (SEM) compression testing.

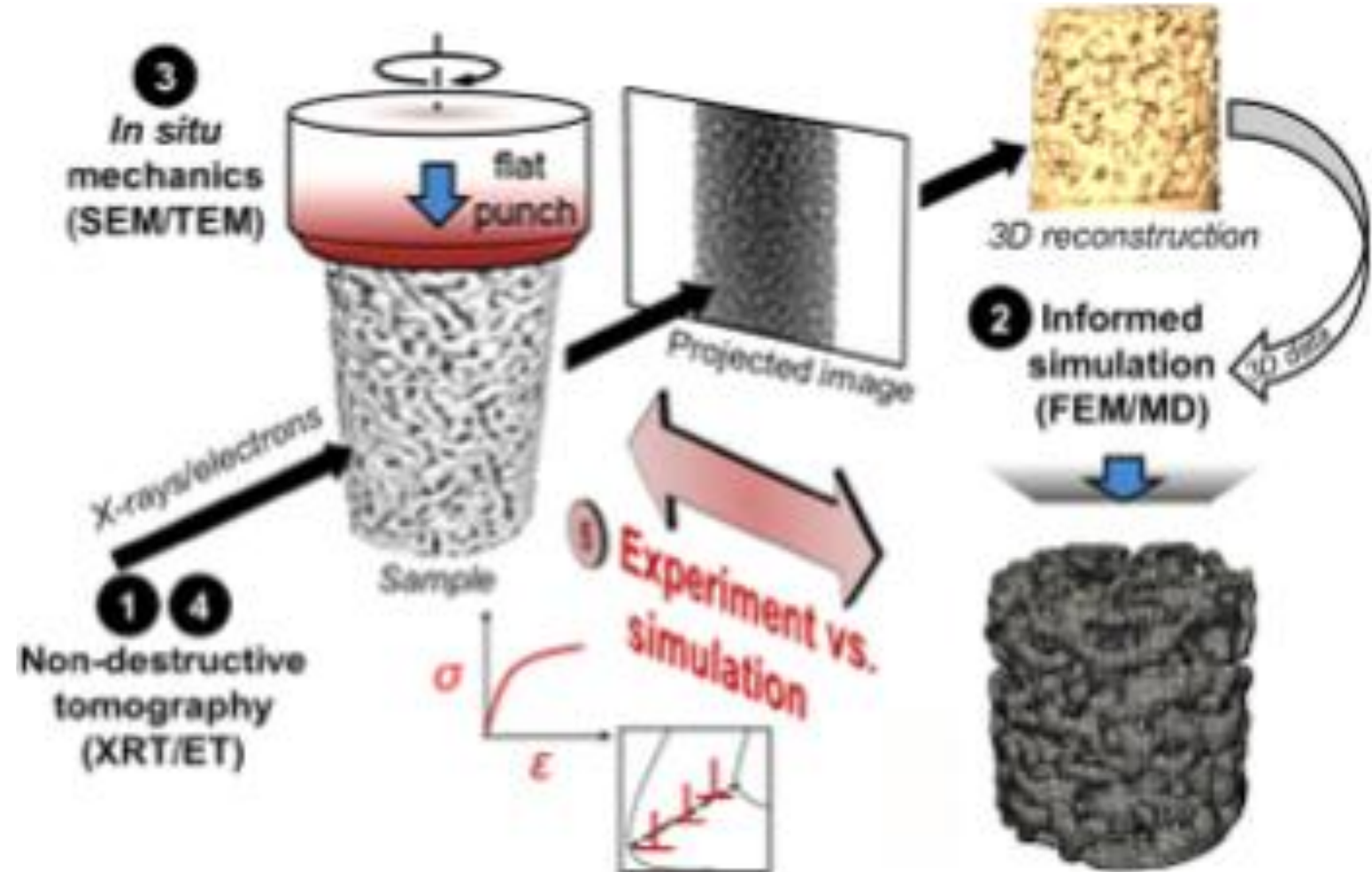


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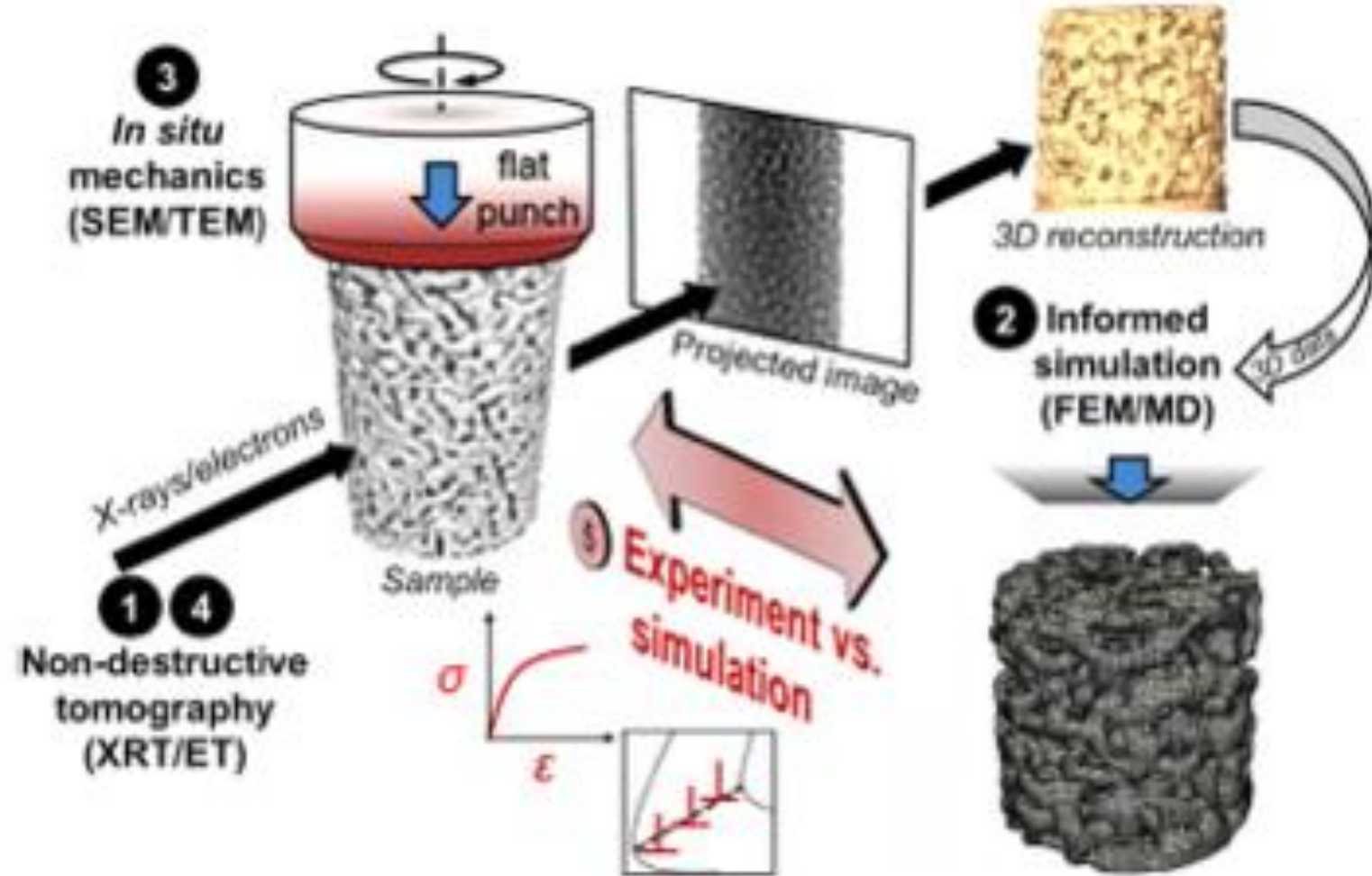




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